Central Tendency I

Simple averages: the arithmetic mean, the median and the mode.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 The idea · types of averages

A measure of central tendency is a **single value that represents the centre or typical value** of a dataset.

Simple averages this chapter	Mean · Median · Mode
Special averages	Geometric mean · Harmonic mean
Positional averages	Quartiles · Deciles · Percentiles

Objectives: condense data into a single value; make comparisons easy.

2 A good average should be...

- ✓ Rigidly defined
- ✓ Simple to understand and easy to calculate
- ✓ Based on **all** values of the data
- ✓ Capable of further mathematical treatment
- ✓ Stable from sample to sample (sampling stability)
- ✓ Not unduly affected by extreme values

No single average meets all six. The mean fails only the last one.

3 Arithmetic mean

The sum of the observations divided by their number. It is what people usually mean by "average".

Raw data

$$\bar{x} = \frac{\sum x_i}{n}$$

Frequency distribution

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

Assumed mean, $d_i = x_i - A$

$$\bar{x} = A + \frac{\sum f_i d_i}{\sum f_i}$$

Coding, $u_i = (x_i - A)/C$

$$\bar{x} = A + C\,\frac{\sum f_i u_i}{\sum f_i}$$

Weighted mean

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

Combined mean of groups

$$\bar{x} = \frac{\sum n_j \bar{x}_j}{\sum n_j}$$

For grouped data, x_i is the class mark, so the mean is an approximation. With equal weights, the weighted mean is the simple mean.

4 Properties of the mean

Deviations sum to zero

$$\sum (x_i - \bar{x}) = 0$$

Least squares

$$\sum (x_i - a)^2 \text{ is least at } a = \bar{x}$$

Change of origin & scale

$$y_i = a + bx_i \Rightarrow \bar{y} = a + b\bar{x}$$

Combine group means by weighting with group sizes, not by averaging the means.

5 Median

The **middle value** after arranging the data in ascending order. It is a **positional** average.

Odd n

$\left(\frac{n+1}{2}\right)^{\text{th}}$ observation

Even n

mean of the $\left(\frac{n}{2}\right)^{\text{th}}$ and $\left(\frac{n}{2}+1\right)^{\text{th}}$ observations

Grouped data (linear interpolation)

$$Md = L + \left(\frac{\frac{N}{2} - F}{f_m}\right) C$$

Median class holds the $(N/2)^{\text{th}}$ value · L its lower boundary · F cumulative frequency below L · f_m its frequency · C its width. Graphically, read it off the less-than ogive at $N/2$.

6 Mode

The value that **occurs most often**. Found by inspection for raw data. It is the right average for **qualitative** (nominal) data.

Grouped data (equal class widths)

$$Mo = L + \left(\frac{f_m - f_p}{2f_m - f_p - f_s}\right) C$$

Modal class has the highest frequency · L its lower boundary · f_p, f_s frequencies of the preceding and succeeding classes · C class width. Graphically, use the histogram.

With **unequal widths**, compare frequency densities rather than raw frequencies. Data can be **bimodal** or have no mode.

7 Mean vs median vs mode

	Mean	Median	Mode
Based on	All values (calculated)	Position	Frequency
Uses every observation	Yes	No	No
Extreme values	Affected very much	Not influenced	Not affected
Open-end classes	Cannot compute	If median class is known	Modal category, yes
From a graph	No	Ogive	Histogram
Qualitative data	No	Ordinal only	Yes
Always unique	Yes	Yes	May be several or none

8 Shape & order

- **Symmetric**, unimodal: mean = median = mode
- Many **positively skewed** distributions: mean > median > mode
- Many **negatively skewed** distributions: mean < median < mode

The skewed orderings are typical, **not** a universal theorem.

Positive observations (preview)

$$AM \geq GM \geq HM$$

9 Worked results from the chapter

Problem	Working	Answer
Mean of 2, 4, 7, 8, 11, 12	$44/6$	**7.33**
Weighted mean, x = 10, 20, 30; w = 1, 2, 3	$(10 + 40 + 90)/6$	**23.33**
Median height, N = 60, median class 160–165	$160 + \frac{30-28}{18} \times 5$	**160.56 cm**
Mode of sorghum ear-head weight, modal class 100–120	$100 + \frac{45-38}{90-38-35} \times 20$	**108.24 g**

! Common mistakes

- × Averaging group means without weighting by group size.
- × Using class **limits** instead of **boundaries** for L.
- × Forgetting to sort the data before finding the median.
- × Applying the mode formula to unequal class widths.

? Fill in the blanks

1. The mean yield of 5 plots is 12 q, so their total yield is ——— q. [60]
2. If 3 cm is added to every plant height, the mean height increases by ——— cm. [3]
3. The deviations of 4, 6 and 8 from their mean add up to ———. [0]
4. For 7 observations arranged in order, the median is the ——— observation. [4th]
5. For grain colour recorded as red, white or brown, the only meaningful average is the ———. [mode]
6. Group A (10 plots, mean 20 q) and group B (30 plots, mean 40 q) have a combined mean of ——— q. [35]

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